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Systematic Review

The Impact of Game-Based Learning on Motivation, Self-Efficacy, and Academic Achievement in the Natural Sciences: A Meta-Analysis

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Abstract

Game-based learning has become an increasingly popular educational methodology due to its ability to enhance student interest and engagement. The aim of this study was to analyze the effect of game-based learning on motivation, self-efficacy, and academic performance in Natural Sciences learning. A systematic review and meta-analytic methodology was employed, following PRISMA guidelines. For this purpose, the databases consulted were Web of Science and Scopus, from which a total of 234 documents were retrieved and reduced to 15 studies after rigorously applying the established eligibility criteria. These studies were included in the systematic review and meta-analysis to ensure the validity and relevance of the meta-analytic findings. The meta-analytic results revealed a very strong and highly significant positive effect across all subgroups, benefiting the experimental groups ($Z = 6.29$; $p < 0.00001$). In conclusion, the implementation of game-based learning has a positive impact on motivation, self-efficacy, and academic performance in the teaching and learning of Natural Sciences content. Therefore, its incorporation into pedagogical practices represents an opportunity to strengthen student engagement and promote more meaningful learning.

Keywords: game-based learning; self-efficacy; natural sciences; motivation; academic achievement

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1. Introduction

Game-based learning (GBL) refers to a type of learning environment that incorporates game content and dynamics due to its ability to increase student engagement, promote active learning (Hubahib & Walag, 2025), and enhance the acquisition of knowledge and skills (Qian & Clark, 2016). Generally, this type of learning includes playful activities related to problem-solving and presents challenges to students, providing them with a sense of achievement (Hu et al., 2024). Well-designed educational games show great potential to improve teaching and learning (Y. Chen & Law, 2021). In science education, GBL is an effective method to support students' learning of scientific knowledge (Safitri et al., 2025). Previous studies have demonstrated that this type of learning not only positively affects students' learning but can also improve their self-efficacy (Czok & Weitzel, 2025; Hung et al., 2014). Consequently, an increasing number of researchers and educators are integrating GBL into science classrooms.

Currently, there is growing interest in STEAM education (Science, Technology, Engineering, Arts, and Mathematics), particularly in science teaching (Zhusupkalieva et al., 2025). The field of Natural Sciences encompasses content from astronomy, biology, physics,

geology, and chemistry (Jasso et al., 2021), and its teaching relies on various instructional models, including transmission–reception, discovery, meaningful reception, conceptual change, and inquiry (Ruiz, 2007). Science is understood as the exploration of the natural world and the explanation of its phenomena and objects (Yager, 2015), while scientific literacy is considered an essential competence for all students (Gao et al., 2016). In this context, GBL emerges as a method with great potential to transform science teaching and learning. Its main characteristic is the integration of games into teaching to promote the acquisition of knowledge and skills (Qian & Clark, 2016). During gameplay, students face problems and challenges linked to the course content, facilitating immersion and active learning. Previous studies have reported positive effects of GBL in science education (C. H. Lin et al., 2013), showing that students experience flow (a psychological state of intense concentration, enjoyment, and loss of time awareness) accompanied by positive affect, high empathy, and behavioral engagement. Consequently, games and playful experiences represent effective tools to enhance learning in science (Badajos et al., 2023).

In general, researchers have examined two types of games: digital and non-digital (Hou et al., 2023). A digital game refers to a game played using electronic devices such as computers, mobile devices, or other electronic devices. Previous studies have found that digital games are effective in enhancing students' learning of scientific knowledge. For example, Sung and Hwang (2013) conducted an experiment integrating educational computer games into elementary science courses and found that students' academic performance improved. Non-digital games, on the other hand, refer to any type of board game, card game, or other games played in physical environments (Uzun et al., 2013). Unlike digital games, non-digital games do not require electronic devices and can be implemented in various settings. They are widely used in STEM education and have been shown to be effective in improving learning outcomes (Liu & Chen, 2013).

Motivation, understood as a psychological process that drives, directs, and sustains behavior toward goal achievement, can be intrinsic—when the action is performed for interest, satisfaction, or enjoyment—or extrinsic—when guided by external rewards (Deci & Ryan, 1985). Motivation plays a key role in student persistence, effort, and engagement in academic tasks (Ahmed et al., 2023) and is closely linked to self-efficacy, defined as the belief in one's ability to organize and execute the actions necessary to achieve specific goals (Bandura, 1982, 1994). Self-efficacy influences how students regulate their emotions, effort, motivation, and behavior during learning; students who perceive themselves as capable of facing challenges are more motivated to engage and persist in the face of difficulties (Liem, 2022).

Therefore, an effective educational game should incorporate appropriate challenges that reinforce students' perception of self-efficacy (Johnson & Mayer, 2010). Empirical evidence confirms that self-efficacy and motivation are positively related to academic performance (Lu & Lien, 2019). Students with high self-efficacy set more ambitious goals, devote more time and effort, and achieve better results (Sander & Hasselhorn, 2024). Consequently, integrating playful activities in the classroom that strengthen both motivation and self-efficacy leads to significant improvements in academic performance. GBL not only facilitates content acquisition but also enhances students' confidence in their ability to learn and succeed (Zhao et al., 2025).

Nevertheless, it is essential to properly balance the value attributed to scientific knowledge to avoid an excessive process of "scientification". In the educational field, this risk involves replacing the formative and humanistic essence of education with a logic governed exclusively by the criteria and procedures of the empirical sciences. To prevent such distortions, it is crucial that contributions from the Natural Sciences translate into "socially, culturally, and territorially relevant learning experiences for all students" (Villarroel et al., 2022). In this sense, and despite the growing interest in GBL and the promising results

reported—particularly regarding motivation, self-efficacy, and academic performance in Natural Sciences learning—the available evidence remains fragmented and inconclusive. A substantial portion of the literature relies on single-group research designs (Kuo et al., 2024; Li et al., 2022; Yeo et al., 2022), which restrict the ability to establish causal relationships and limit the generalizability of findings. Therefore, the specific analysis of studies comparing experimental groups (EGs) and control groups (CGs) becomes essential to generate more robust evidence and reduce potential biases in interpreting results (Higgins et al., 2022). In this regard, the present study conducts a systematic review and meta-analysis that integrates the available findings, estimates the effect size of game-based interventions, and examines the predominant conceptual connections in the literature, thereby contributing to clarifying the current state of knowledge in this field.

In sum, GBL has emerged as an innovative pedagogical methodology aimed at increasing student engagement and interest, particularly in areas perceived as complex or abstract, such as the Natural Sciences (Hamari et al., 2016). Nevertheless, despite the proliferation of studies on gamification and GBL, findings regarding their effectiveness in enhancing motivation, self-efficacy, and academic achievement remain variable and inconsistent (Sailer & Homner, 2020). Moreover, despite the growing number of systematic reviews and meta-analyses on GBL, significant gaps persist in the literature, as many previous reviews have addressed GBL in a general manner or have focused exclusively on digital contexts, specific educational levels, or isolated variables, without providing an integrated analysis of its effects on motivation, self-efficacy, and academic achievement within the domain of Natural Sciences (Clark et al., 2016; Subhash & Cudney, 2018). In addition, a substantial proportion of prior empirical studies rely on single-group research designs or lack systematic comparisons between an EG and CG, which limits the robustness of the conclusions and the precise estimation of effect sizes (Sitzmann, 2011; Wouters et al., 2013).

The Present Study

This study allows for the consolidation of existing evidence, the identification of patterns, and the quantitative evaluation of the impact of GBL across diverse educational contexts. Its significance lies in addressing these limitations through a systematic review and meta-analysis focused specifically on comparative studies involving an EG and CG, integrating both quantitative and conceptual evidence to provide a more robust and up-to-date understanding of GBL's impact on learning in the Natural Sciences. In doing so, it offers educators and instructional designers evidence-based insights to guide the implementation of more effective teaching strategies.

This study shows that game-based interventions not only significantly enhance students' motivation and self-efficacy but also have a positive impact on academic performance, systematically integrating both quantitative and conceptual findings. In addition, it identifies recurring patterns and thematic connections in the literature on GBL in the Natural Sciences, providing a stronger foundation for future research and educational practice. Indeed, by rigorously comparing the effects of an EG and CG, this study refines and extends the results of previous meta-analyses, showing that the positive effects of GBL are generally consistent but may vary depending on the psychological domain studied and/or the educational context. Furthermore, it challenges prior conclusions by integrating conceptual evidence that allows for a better understanding of the mechanisms underlying GBL's effects, offering a more comprehensive and nuanced view of its impact on learning in the Natural Sciences.

The objective of this study is to analyze the effect of GBL on motivation, self-efficacy, and academic performance in Natural Sciences learning, and based on this objective, the following research questions are posed:

(a) What is the average effect size of game-based interventions on motivation, self-efficacy, and academic performance when comparing EG with CG?
(b) Which conceptual and thematic connections most frequently emerge in the literature on GBL applied to the Natural Sciences?

2. Materials and Methods

2.1. Procedure

This study is grounded in a systematic review of the scientific literature. Its development followed the guidelines established by the PRISMA statement (Page et al., 2021), which ensured the transparency and reproducibility of the procedures employed (García-Peñalvo, 2022).

2.2. Eligibility Criteria

First, the following inclusion criteria were established: (a) empirical studies with no restrictions on publication date; (b) research including two comparable groups (experimental and control); (c) studies conducted either face-to-face or online; (d) studies in which the variables measured were motivation, self-efficacy, and academic performance in the learning of Natural Sciences; (e) research conducted at any educational level; (f) studies with samples that included students with and without special educational needs; and (g) articles on GBL, both digital and non-digital.

Likewise, regarding the type and completeness of outcome data, only studies that provided sufficient statistical information to calculate effect sizes were included. Specifically, studies were required to report, at a minimum, means and standard deviations and sample sizes for the EG and CG, or equivalent statistics that allowed for the estimation of effect sizes. Studies that did not provide sufficient quantitative data were excluded from the meta-analysis.

On the other hand, the following exclusion criteria were defined: (a) duplicate studies; (b) studies with restricted access to the publication; (c) book chapters, books, or conference proceedings; (d) review articles, theoretical studies, cross-sectional research, descriptive studies, or case studies; (e) research written in languages other than Spanish or English; (f) studies irrelevant to the research topic (e.g., those analyzing GBL with variables other than those considered in this study); (g) research consisting of only a single experimental group; (h) studies lacking sufficient information for meta-analysis; (i) research examining the influence of GBL on motivation, self-efficacy, and academic performance together with other methodologies (e.g., inquiry-based learning).

After the selection process, the included studies were grouped according to the variables under analysis (motivation, self-efficacy, and academic performance), as well as by the type of GBL implementation (digital or non-digital). This classification allowed the information to be organized coherently for both the qualitative and quantitative synthesis.

2.3. Information Sources

The search was conducted without temporal restrictions in the Web of Science (WoS) and Scopus databases, both part of the Elsevier group and recognized for their prestige, as well as for including the JCR (*Journal Citation Reports*) and SJR (*SCImago Journal Rank*) impact indices, respectively. According to the recommendations of Pranckutė (2021), these databases are considered highly effective tools for conducting systematic reviews. The search was carried out in November 2025.

2.4. Search Strategies

The search was conducted in the WoS and Scopus databases. The search formula used in both databases was: ((“game-based learning”) AND (“motivation” OR “self-efficacy”

OR "academic performance") AND ("natural sciences" OR "biology" OR "physics" OR "chemistry" OR "geology" OR "astronomy")).

No temporal restrictions were applied, and no additional filters (e.g., language or document type) were used. This strategy allowed for a comprehensive and reproducible identification of relevant studies in both databases.

2.5. Study Selection Process

Rigorous strategies were implemented for study selection, including a clear definition of inclusion criteria, the systematic application of search terms across multiple databases, and manual reference review. The selection process was carried out in several stages:

(1) Initial screening by title and abstract: All identified records were evaluated by the three researchers.
(2) Full-text assessment: Preselected articles were read in their entirety to apply the inclusion and exclusion criteria.
(3) Duplicate detection: An Excel file was used to organize the studies and identify duplicates.

Throughout the process, the titles and abstracts of all identified records were carefully reviewed, and preselected records were assessed in full text. Information from studies that met the eligibility criteria was extracted into an Excel file.

The search strategy yielded a total of 234 documents, with 136 from WoS and 98 from Scopus. Regarding the exclusion criteria, a total of 219 documents were discarded according to the established criteria: 48 studies were excluded based on criterion (a); 4 based on criterion (b); 74 based on criterion (c); 21 based on criterion (d); 2 based on criterion (e) (one published in Chinese and another in Turkish); 53 based on criterion (f); 15 based on criterion (g); and 2 based on criterion (h).

2.6. Data Extraction Process

For data extraction, a structured data extraction form was designed based on the previously established inclusion criteria. In the first stage, the following information was coded: (1) author(s); (2) year of publication; (3) study objective. Subsequently, the following fields were completed: (4) study; (5) language of publication; (6) location; (7) study design; (8) number of participants per group; (9) age or mean age; (10) variables analyzed. Finally, a third questionnaire was completed, including: (11) study; (12) educational level; (13) instruments used to assess motivation, self-efficacy, and academic performance; (14) duration of the intervention; (15) type of game.

The most relevant qualitative and quantitative data were organized in an Excel spreadsheet. The process was carried out independently by three researchers (JGSS, DAR, and RQL) to ensure accuracy and reliability. Any discrepancies were resolved through a second reading of the documents until consensus was reached. No additional automated tools were used, nor were the original study authors contacted.

Finally, the results were summarized in a table presenting the key characteristics of the included studies, in accordance with the PRISMA statement recommendations.

2.7. List of Data: Outcomes and Other Variables

All outcomes of interest for this study were identified and defined, focusing on motivation, self-efficacy, and academic performance in GBL in the Natural Sciences. All compatible results for each outcome were collected, including instruments used to measure the variables and study designs. Additionally, other relevant variables were recorded, such as participant characteristics (age, sex, educational level) and intervention characteristics (type of game, duration, and mode of implementation). In cases where information was missing or uncertain, clear assumptions were established based on the available data

and a thorough review of the studies, ensuring consistency and transparency in the data extraction process. For example, when the exact age of the participants was not specified, it was estimated based on the average age range reported in comparable studies.

2.8. Data Analysis: Meta-Analysis

A review of interventions was conducted to evaluate their effectiveness, estimating the effect size using the mean difference (*MD*) with 95% confidence intervals (CIs), which allowed for quantifying the magnitude of change and comparing results across studies using similar measures (Sánchez-Meca & Ato, 1989). The included studies were selected based on predefined criteria for population, intervention, comparison, and outcome (PICO), and their characteristics were tabulated to determine eligibility for each synthesis. The data were prepared by reviewing missing values and applying conversions or adjustments when necessary to ensure comparability across studies.

Individual study results were organized into tables and forest plots using RevMan, facilitating visualization of effect sizes, CIs, and heterogeneity (Sánchez-Meca & Ato, 1989). The meta-analysis was conducted using a random-effects model, as this approach accounts for variability among included studies and assumes that observed differences in effect sizes are not due solely to chance, but also to study-specific or contextual factors (Borenstein et al., 2010). This approach maximizes statistical power and reduces the influence of random variability between studies, providing a more consistent and robust interpretation of the results.

The direction of effect sizes was considered favorable when results indicated improvement in the interventions groups. Statistical significance was assessed with $p < 0.05$. Heterogeneity was measured using the I^2 statistic, which expresses the percentage of variability across studies (Higgins et al., 2003; Bölek et al., 2021), interpreted as low (0–25%), low to moderate (26–50%), moderate to high (51–75%), or high (>75%). High heterogeneity does not invalidate the meta-analysis but requires cautious interpretation and exploration of potential sources of discrepancy.

Finally, sensitivity analyses were conducted to evaluate the robustness of the results, checking whether the conclusions remained consistent under possible variations in the included studies or analysis methods. These analyses are described in Section 3.5, and they confirmed that the main conclusions remained consistent: the effect sizes and the direction of the results did not change significantly, supporting the robustness of our findings.

2.9. Risk of Bias Assessment

The risk of bias in the studies included in the meta-analysis was assessed using *Review Manager* (RevMan) software, Cochrane, version 5.4 (Cochrane, London, United Kingdom), following the guidelines of Higgins et al. (2003). The assessment was conducted independently by two researchers (JGS-S and RQ-L), with any discrepancies resolved through discussion and consensus with a third researcher (DA-R).

For the evaluation of publication bias, funnel plots were generated in RevMan and the distribution of points was examined; a symmetrical distribution around the overall effect was considered indicative of a low risk of publication bias, whereas an asymmetrical distribution suggested the potential presence of bias. This approach allowed for a systematic, rigorous, and transparent assessment of the risk of bias across all included studies.

2.10. Assessment of the Certainty of Evidence

The certainty or confidence in the evidence for each outcome was assessed using the GRADE (Grading of Recommendations, Assessment, Development and Evaluation) approach. This method considers factors such as risk of bias, inconsistency of results, imprecision, indirectness of evidence, and risk of publication bias. Each outcome was rated

in terms of certainty as high, moderate, low, or very low, according to the magnitude of these factors.

The assessment was conducted independently by two researchers (JGS-S and RQ-L), with any discrepancies resolved through discussion and consensus with a third researcher (DA-R). This procedure ensured a systematic, transparent, and consistent evaluation of the confidence in the results of the included studies.

2.11. Co-Occurrence Analysis

With the aim of analyzing the co-occurrence of keywords derived from the literature search, an automated analysis was conducted using *VOSviewer* version 1.6.20, a tool specialized in the construction and visualization of bibliometric networks (Van Eck & Waltman, 2010). The resulting keyword groups were interpreted qualitatively to identify the main thematic areas within the field of study.

Each cluster was evaluated considering its thematic relevance and its degree of connection with other groups within the scientific literature, which allowed for a more precise understanding of prevailing trends and research focal points in the area.

3. Results

The established eligibility criteria resulted in the inclusion of 15 studies for review. The selection process is summarized in the PRISMA flow diagram (Figure 1).

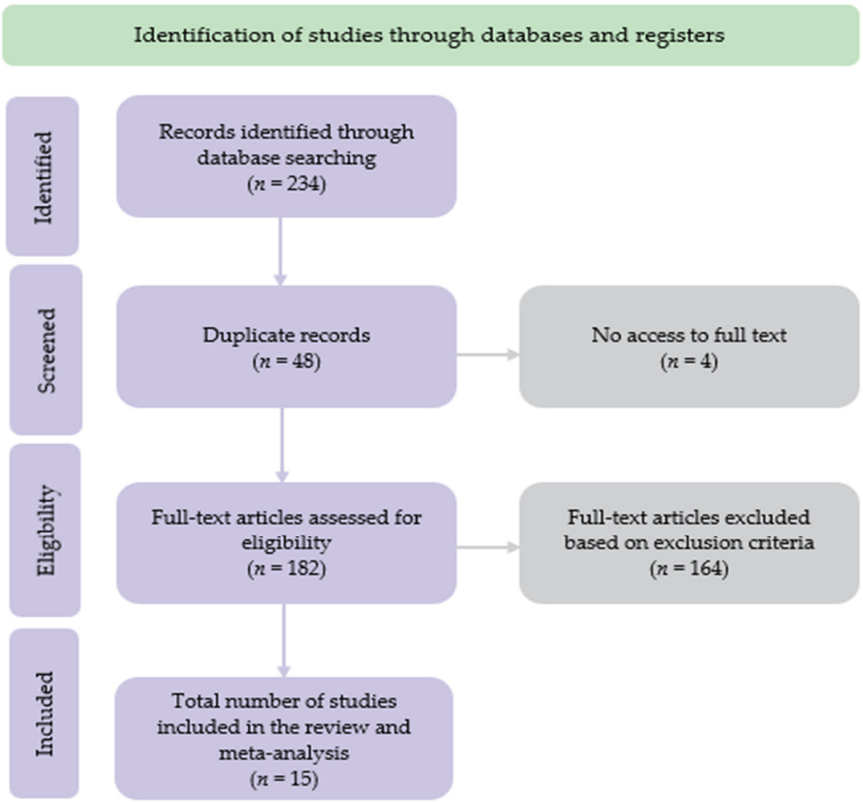

Figure 1. Screening of articles for the systematic review according to the PRISMA flow.

3.1. Descriptive Analysis of the Selected Studies

The studies included in this review cover publications from 2012 to 2025 (Cha et al., 2017; Chu & Chang, 2014; Hwang et al., 2012, 2013, 2024; Khamparia et al., 2020; Wang & Zheng, 2021; Rahimi et al., 2022; K. F. Chen et al., 2024; García & Garzón, 2024; Lazarevic et al., 2024; Mohammed & Karpudewan, 2024; Zhao et al., 2025), reflecting growing interest in educational games in the Natural Sciences. Table 1 summarizes the objectives of the

studies, all of which aimed to evaluate how digital or physical GBL strategies improve students' motivation, self-efficacy, and academic performance, regardless of differences in game design or discipline.

Most studies employed quasi-experimental or pre-test/post-test designs, with sample sizes ranging from 24 to 693 participants aged 10–17 (Cha et al., 2017; K. F. Chen et al., 2024; Chu & Chang, 2014; García & Garzón, 2024; Hwang et al., 2012, 2013, 2024; Khamparia et al., 2020; Lazarevic et al., 2024; Mohammed & Karpudewan, 2024; Rahimi et al., 2022; Sung & Hwang, 2013; Wang & Zheng, 2021; Yang & Chang, 2013; Zhao et al., 2025). The studies were conducted in various countries, including Taiwan, Malaysia, Colombia, China, the United States, and Serbia.

The main variables analyzed included learning and academic performance, understanding of scientific content, knowledge retention, critical thinking, problem-solving, motivation, self-efficacy, attitudes toward learning, satisfaction, perception of competence, active participation, technological acceptance, and cognitive load (Cha et al., 2017; K. F. Chen et al., 2024; Chu & Chang, 2014; García & Garzón, 2024; Hwang et al., 2012, 2013, 2024; Khamparia et al., 2020; Lazarevic et al., 2024; Mohammed & Karpudewan, 2024; Rahimi et al., 2022; Sung & Hwang, 2013; Wang & Zheng, 2021; Yang & Chang, 2013; Zhao et al., 2025) (Table 2).

Studies covered primary, secondary, and higher education levels. Primary-level studies included Hwang et al. (2012, 2013), Chu and Chang (2014), Khamparia et al. (2020), Sung and Hwang (2013), and Lazarevic et al. (2024). Secondary-level studies included K. F. Chen et al. (2024), García and Garzón (2024), Hwang et al. (2024), Mohammed and Karpudewan (2024), Rahimi et al. (2022), Wang and Zheng (2021), and Zhao et al. (2025). Higher education was represented by Cha et al. (2017).

Various instruments were used to measure motivation, self-efficacy, and academic performance. For example, Cha et al. (2017) used the ACSIv2 questionnaire (Bauer, 2008) and final exam scores. García and Garzón (2024) applied Keller's (2010) IMMS, Rahimi et al. (2022) used the Intrinsic Motivation Inventory (IMI; Ryan, 1982), Hwang et al. (2024) assessed scientific self-efficacy using P. R. Pintrich et al.'s (1991) questionnaire, and Wang and Zheng (2021) measured general self-efficacy with the NGSE scale. Academic performance was typically assessed via pre- and post-tests, final exams, or transfer tests (Hwang et al., 2012, 2013; Chu & Chang, 2014; Cha et al., 2017; García & Garzón, 2024; Khamparia et al., 2020; Yang & Chang, 2013; Zhao et al., 2025; Rahimi et al., 2022).

Intervention durations varied from single sessions of 50–120 min (Hwang et al., 2012, 2013; Chu & Chang, 2014; Sung & Hwang, 2013; Wang & Zheng, 2021) to multi-week programs (K. F. Chen et al., 2024; Lazarevic et al., 2024; Yang & Chang, 2013; Rahimi et al., 2022). Some studies did not fully specify duration but reported clear pre- and post-intervention testing periods (García & Garzón, 2024; Mohammed & Karpudewan, 2024).

The types of games used were diverse. Puzzle or logic games were applied by Cha et al. (2017) and Wang and Zheng (2021); educational role-playing games by Hwang et al. (2012, 2013) and Chu and Chang (2014); concept map-based digital games by K. F. Chen et al. (2024); mission-structured digital games by García and Garzón (2024); collaborative knowledge-organizer games by Zhao et al. (2025); 3D digital microworlds by Khamparia et al. (2020); and board games like STEM-PT Traveler by Mohammed and Karpudewan (2024). Game creation as a learning tool was explored by Yang and Chang (2013) (Table 3).

Table 1. Qualitative summary of the studies: author(s), year of publication, and objective.

Author(s) (Year of Publication)	Objective
Cha et al. (2017)	To incorporate a brainteaser game into a basic organic chemistry course to improve students' attitudes and academic performance
K. F. Chen et al. (2024)	To investigate the impacts of knowledge-checking based on concept mapping (KC-CMG) in digital games
Chu and Chang (2014)	To propose and evaluate a two-tier test approach for educational video games to improve learning and motivation in identifying migratory birds
García and Garzón (2024)	To evaluate how a video game based on cognitive multimedia theory and game-based learning affects learning, retention, and motivation of ninth-grade students in natural sciences
Hwang et al. (2012)	To develop a personalized educational game based on students' learning styles
Hwang et al. (2013)	To evaluate a digital game approach with integrated concept maps to improve learning, reduce cognitive load, and increase acceptance and motivation of primary students in natural sciences compared to conventional digital games
Hwang et al. (2024)	To evaluate the effectiveness of a digital game model integrating concept maps and collaborative conceptual planning (CM-DGBL-CCP) in problem-solving in chemistry compared to T-DGBL-CCP
Khamparia et al. (2020)	To apply microworld game-based learning to students with neuromuscular disabilities to improve learning, motivation, self-efficacy, and reduce cognitive load compared to the conventional approach
Lazarevic et al. (2024)	To apply the Nikola Tesla Center learning system to assess its impact on student motivation when learning abstract physics concepts using a card-based learning approach
Mohammed and Karpudewan (2024)	To improve students' motivation toward learning chemistry, particularly about the Periodic Table
Rahimi et al. (2022)	To investigate how the timing of learning supports (before or after playing) in educational games impacts student outcomes in physics
Sung and Hwang (2013)	To evaluate the effectiveness of a collaborative game-based learning approach integrating a Mindtool (repertory grid) to improve attitude, motivation, self-efficacy, and learning achievements in natural science courses
Wang and Zheng (2021)	To use game-based learning (digital and non-digital) to support science learning (reflection of light)
Yang and Chang (2013)	To compare the effectiveness of digital game authoring and a comparison group (Flash design) on focus, critical thinking, and academic performance
Zhao et al. (2025)	To determine whether a digital game based on collaborative knowledge organizers (CKO-DG) results in higher performance and lower cognitive load in biology compared to KO-DG and CDG

Note. CDG = Conventional digital game, without emphasis on knowledge organizers; CM-DGBL-CCP = Digital game-based learning approach that integrates concept maps (CM) to organize and relate ideas, digital game-based learning (DGBL), and collaborative conceptual planning (CCP); CKO-DG = Digital game using collaborative knowledge organizers; KC-CMG = Knowledge Construction-Concept Map Generation; KO-DG = Digital game with individual or non-collaborative knowledge organizers; T-DGBL-CCP = Similar model without concept maps, used to evaluate whether including concept maps improves learning outcomes.

Table 2. Qualitative summary of the studies: study, publication language, location, design, participants, age, and analyzed variables.

Study	Publication Language	Location	Study Design	Participants in Each Group (*n*)	Age/Mean Age (M_{age})	Variables Analyzed
Cha et al. (2017)	English	Malaysia	Quasi-experimental	EG n = 98 CG n = 100	Not specified	Student attitudes (intellectual accessibility and emotional satisfaction) and academic achievement
K. F. Chen et al. (2024)	English	Taiwan	Pre-test and post-test design	EG n = 24 CG n = 24	Not specified	Learning achievement, problem-solving awareness, tendency to critical thinking, intrinsic cognitive load, and learning behavioural patterns
Chu and Chang (2014)	English	Taiwan	Quasi-experimental	EG n = 26 CG n = 27	M_{age} = 11.80	Learning achievement, learning motivation, and technology acceptance (perceived ease of use and perceived usefulness)
García and Garzón (2024)	Spanish	Colombia	Quantitative approach, empirical analytical paradigm, experimental design	EG n = 29 CG n = 27	Between 13 and 15 years old	Learning, retention, and motivation (attention, relevance, confidence, satisfaction)
Hwang et al. (2012)	English	Taiwan	Quasi-experimental	EG n = 24 CG n = 22	Between 10 and 11 years old	Learning performance, learning motivation, technology acceptance (perceived ease of use and perceived usefulness)
Hwang et al. (2013)	English	Taiwan	Quasi-experimental	EG n = 28 CG n = 28	Between 11 and 12 years old	Learning achievement, learning motivation, cognitive load (mental load and mental effort), technology acceptance (perceived ease of use and perceived usefulness)
Hwang et al. (2024)	English	Taiwan	Quasi-experimental	EG n = 49 CG n = 56	Between 14 and 15 years old	Problem-solving tendency, learning achievement in chemistry, scientific self-efficacy, scientific learning strategies, cognitive load, learning behavioural patterns
Khamparia et al. (2020)	English	Asia	Quasi-experimental	EG n = 30 CG n = 30	Not specified	Personality level, anxiety level, cognitive load, learning motivation, self-efficacy, and learning achievement
Lazarevic et al. (2024)	English	Serbia	Quasi-experimental	EG n = 693 CG n = 678	Not specified	Motivation, engagement, and academic performance

Table 2. *Cont.*

Study	Publication Language	Location	Study Design	Participants in Each Group (n)	Age/Mean Age (M_{age})	Variables Analyzed
Mohammed and Karpudewan (2024)	English	Malaysia	Quasi-experimental	EG n = 46 CG n = 45	Between 16 and 17 years old	Motivation, engagement, and academic performance
Rahimi et al. (2022)	English	United States	Pre-test/post-test between subjects with random assignment to 3 conditions	EG n = 46 CG n = 53	M_{age} = 14.33	Physics understanding (near and far transfer), subjective perceptions (enjoyment, competence, effort, frustration, value), and game variables (level duration, solution rate)
Sung and Hwang (2013)	English	Taiwan	Quasi-experimental with 3 groups: Experimental (GBL); Control A (not GBL); and CG	EG n = 31 CG A n = 31 CG B n = 31	Between 11 and 12 years old	Learning achievement, learning attitudes, learning motivation, and group learning self-efficacy
Wang and Zheng (2021)	English	China	Quasi-experimental (3 groups: digital, non-digital, control)	EG n = 31 CG n = 32	M_{age} = 12.60	Scientific content knowledge achievement, and self-efficacy
Yang and Chang (2013)	English	Taiwan	Quasi-experimental	EG n = 32 CG n = 35	Between 13 and 14 years old	Concentration, critical thinking skills, and academic achievement
Zhao et al. (2025)	English	China	Quasi-experimental	EG n = 33 CG n = 28	M_{age} = 14	Learning achievement, cognitive load, and learning motivation

Note. CG = Control Group; EG = Experimental Group; GBL = Game-Based Learning.

Table 3. Qualitative summary of studies: study, educational stage, instruments used, duration, and type of game.

Study	Educational Stage	Assessment Tools Used (Motivation, Self-Efficacy, and Academic Performance)	Duration of Intervention	Type of Game
Cha et al. (2017)	Higher Education	Academic Performance: Final exam (implicit); Motivation/Attitude: ACSIv2 questionnaire (measuring Intellectual Accessibility and Emotional Satisfaction) (Bauer, 2008)	Twice (in weeks 12 and 13) during the last 15 min of the lesson	Brainteaser game
K. F. Chen et al. (2024)	Secondary Education	Academic Performance: Posttest (learning achievement); Motivation: Mention of improved motivation in qualitative interviews	2 weeks (twice a week)	Digital game integrating concept mapping and knowledge checking (KC-CMG)

Table 3. *Cont.*

Study	Educational Stage	Assessment Tools Used (Motivation, Self-Efficacy, and Academic Performance)	Duration of Intervention	Type of Game
Chu and Chang (2014)	Primary Education	Motivation: Motivation Questionnaire by Hwang et al. (2011); Academic Performance: Pre-test and Post-test of learning achievements (multiple choice, matching items, fill in the blanks)	60 min	Role-playing game based on a two-tier test approach for identifying migratory birds
García and Garzón (2024)	Secondary Education	Motivation and Self-Efficacy: IMMS Survey (Keller, 2010), which measures four parameters: Attention, Relevance, Confidence (related to self-efficacy), and Satisfaction; Academic Performance: Pre-test, Post-test, and Follow-up Test (Retention), using ten multiple-choice questions	One class session (exact time not specified)	Educational video game (developed with Genially) based on GBL and CTML theory, with a role-playing structure (missions)
Hwang et al. (2012)	Primary Education	Academic Performance: Pre-test and Post-test; Motivation: Learning Motivation Questionnaire (modified from R. R. Pintrich & DeGroot, 1990); Self-efficacy: Implicit in the motivation/self-regulation components measured	60 min	Personalized educational role-playing game
Hwang et al. (2013)	Primary Education	Motivation: Learning Motivation Questionnaire (Hwang et al., 2011); Academic Performance: Pre-test and Post-test (multiple choice, matching, question-answer)	120 min	Educational role-playing game incorporating concept maps (concept map-embedded gaming approach)
Hwang et al. (2024)	Secondary Education	Academic Performance: Achievement in chemistry; Motivation: Scientific Learning Strategy Questionnaire (Lee et al., 2008); Scientific Self-Efficacy Questionnaire by P. R. Pintrich et al. (1991)	60 min	Digital game (CM-DGBL-CCP)
Khamparia et al. (2020)	Primary Education	Motivation: Motivation questionnaire (scales by McAuley et al., 1989, and Hwang et al., 2011); Self-efficacy: Self-efficacy Questionnaire (Gaumer et al., 2016); Academic performance: Pre-test and post-test using multiple-choice questions/puzzles	60 min	Microworld game-based learning system called "SmartLearn" (3D graphical simulation system)
Lazarevic et al. (2024)	Primary Education	Motivation and Self-efficacy: SMTPL (Students' Motivation Toward Physics Learning) questionnaire, which measures six motivational factors, including self-efficacy, active learning strategies, value of learning physics, performance goal, achievement goal, and stimulation of the learning environment (Tuan et al., 2005)	6 weeks (10 min once a week)	Educational memory card game (NTC4memory) as part of the Nikola Tesla Center learning system (NTC)

Table 3. *Cont.*

Study	Educational Stage	Assessment Tools Used (Motivation, Self-Efficacy, and Academic Performance)	Duration of Intervention	Type of Game
Mohammed and Karpudewan (2024)	Secondary Education	Motivation: Chemical Motivation Questionnaire II (CMQ II) (measures Intrinsic, Career, Grade, Self-determination, and Self-efficacy) (Glynn et al., 2011)	Pre- and post-tests administered one week before and one week after treatment (the exact duration is not specified)	Board game called STEM-PT Traveler
Rahimi et al. (2022)	Middle School and Secondary Education	Academic Performance: Posttest (Near and Far Transfer of Physics); Motivation: Modified Intrinsic Motivation Inventory (IMI) (measures enjoyment, frustration, and value) (Ryan, 1982); Self-efficacy: Perceived Competence (IMI subscale)	Three days (approximately 150 min of individual play)	Educational digital physics game Physics Playground
Sung and Hwang (2013)	Primary Education	Academic Performance: Pre-test and Post-test in natural sciences; Motivation: Learning Motivation Questionnaire (modified from R. R. Pintrich & DeGroot, 1990); Self-efficacy: Questionnaire designed to assess self-efficacy in group learning	100 min	Collaborative educational game assisted by Mindtool (repertory grid)
Wang and Zheng (2021)	Secondary Education	Academic Performance: Scientific Content Knowledge Test (Multiple Choice, Fill-in-the-Blank, and True/False); Self-Efficacy: New General Self-Efficacy Scale (NGSE scale) (G. Chen et al., 2001)	1 session of 50 min	Digital puzzle game (Lazors) and non-digital puzzle game (Laser Maze)
Yang and Chang (2013)	Secondary Education	Academic Performance: Academic achievement (final exam)	19 weeks (game design took place from week 11 to week 15, lasting 45 min per week)	Digital game authoring using RPG Maker software
Zhao et al. (2025)	Secondary Education	Academic Performance: Paper-based test (pre- and post-test) to assess learning achievement; Motivation: Motivation Questionnaire (adapted from P. R. Pintrich et al., 1991)	Two weeks (the exact time is not specified)	Digital game based on collaborative knowledge organizers (CKO-DG)

Note. CKO-DG = Collaborative learning based on structured knowledge with the interactivity and motivation of a digital game; CM-DGBL-CCP = Educational digital game that integrates concept maps and collaboration to enhance students' understanding, retention, and motivation; CTML = Cognitive Theory of Multimedia Learning; IMMS = Instructional Materials Motivation Survey; KC-CMG = Knowledge Construction-Concept Map Generation.

3.2. Synthesis of the Evidence Found

First, at the Primary Education stage, studies indicated that games and knowledge organizers improved academic performance, motivation, and self-efficacy. Chu and Chang (2014) showed that a role-playing game with a two-tier test approach enhanced motivation and learning in the natural sciences. Hwang et al. (2012, 2013) reported that personalized educational games included integrated concept maps, increased motivation, self-efficacy, and academic performance while reducing cognitive load. Similarly, Khamparia et al. (2020) demonstrated that the *SmartLearn* microworld increased motivation, self-efficacy, and learning achievements in students with neuromuscular disabilities. Lazarevic et al. (2024) found that the NTC (Nikola Tesla Center) memory game improved perceived self-efficacy, motivation, and the use of active learning strategies in physics. Sung and Hwang (2013) showed that a collaborative game assisted by *Mindtools* enhanced motivation, self-efficacy, and performance in natural sciences learning.

In Secondary Education, studies focused on digital games, concept maps, and knowledge organizers, showing improvements in academic performance, motivation, and higher-order cognitive skills. García and Garzón (2024) reported that a role-playing educational video game increased learning, retention, and motivation in natural sciences. K. F. Chen et al. (2024) found that the KC-CMG (Knowledge Concept–Concept Mapping Game) strategy, based on concept mapping and knowledge checking, enhanced learning outcomes, critical thinking, and problem-solving awareness. Similarly, Mohammed and Karpudewan (2024) showed that the board game STEM-PT Traveler increased intrinsic and career motivation as well as self-efficacy in chemistry. Hwang et al. (2024) demonstrated that a digital chemistry game improved academic performance, motivation, and scientific self-efficacy. Similarly, in Secondary Education, Zhao et al. (2025) showed that the CKO-DG (Collaborative Knowledge Organizer Digital Game) approach in biology enhanced academic performance, motivation, and collaborative awareness while reducing cognitive load. Other studies, such as Wang and Zheng (2021), indicated that digital and non-digital puzzle games improved scientific knowledge and self-efficacy, whereas Rahimi et al. (2022) reported increases in motivation, self-efficacy, and learning transfer in physics through the digital game *Physics Playground*. Finally, Yang and Chang (2013) demonstrated that digital game authoring using RPG Maker fostered critical thinking and learning retention in biology.

Finally, in higher education, although the evidence was more limited, the incorporation of GBL in the natural sciences also showed benefits. In this regard, Cha et al. (2017) reported that a puzzle game in basic organic chemistry increased positive attitudes toward discipline, improving intellectual accessibility and emotional satisfaction.

3.3. Summary of Meta-Analytic Results

To adequately interpret the results of the meta-analysis, it is important to note that the green squares represent the estimated *MD* for each study, with their area proportional to the weight assigned within the random-effects model (Figure 2). The horizontal lines associated with each estimate indicate the 95% CI, while the black diamond synthesizes the overall combined effect, reflecting the width of its corresponding CI.

Regarding the subgroup analyses, the motivation subgroup included $n = 6$ studies (Chu & Chang, 2014; Hwang et al., 2012; Khamparia et al., 2020; Mohammed & Karpudewan, 2024; Rahimi et al., 2022; Zhao et al., 2025), with 191 participants in the EG and 187 in the CG. The combined effect indicated a MD of 0.75 (95% CI: 0.33–1.17; $Z = 3.50$; $p = 0.0005$), corresponding to a moderate effect. The observed heterogeneity was high ($I^2 = 82\%$).

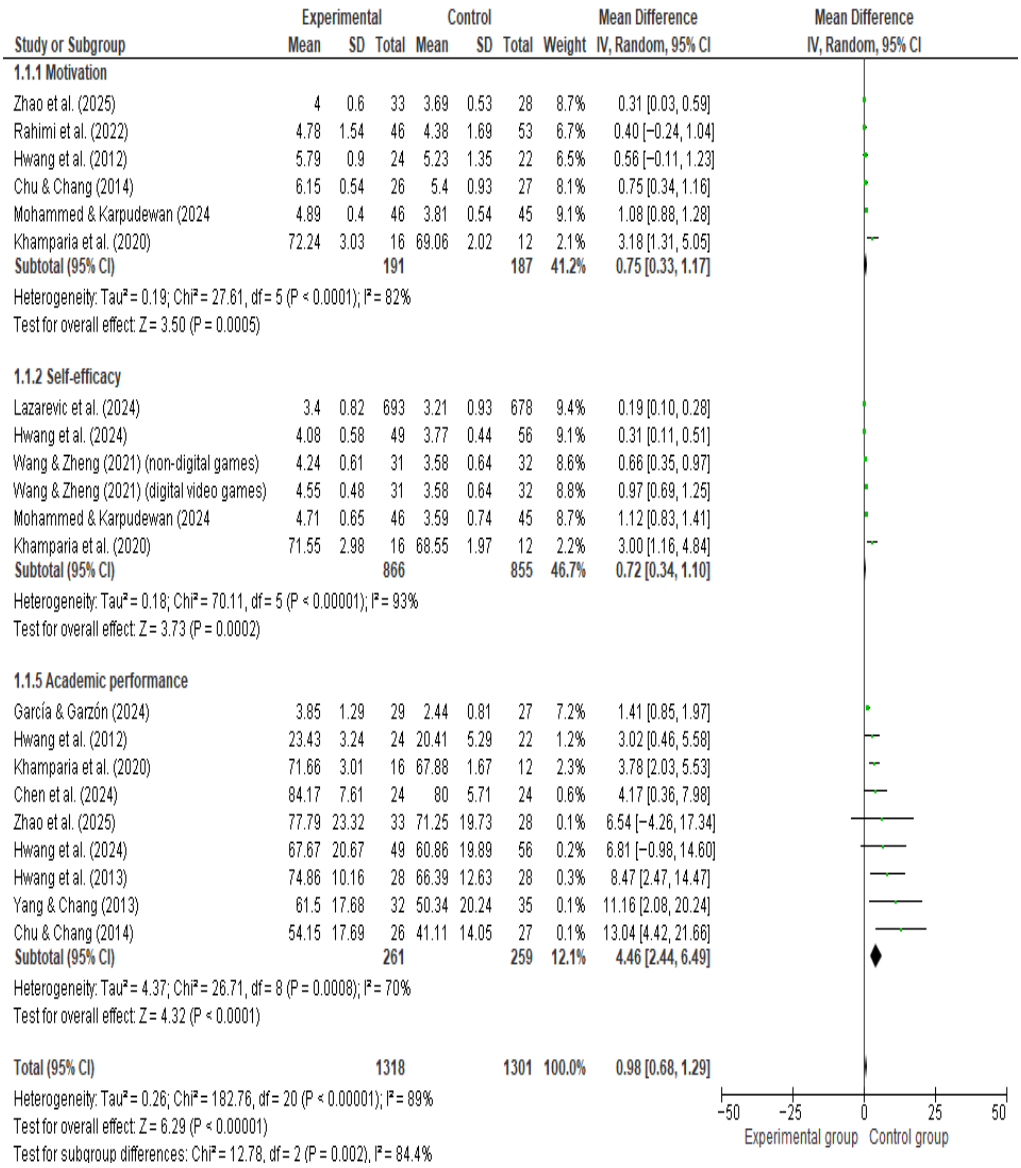

Figure 2. Forest plot of the overall effect size of the interventions. *Note*. Tau^2 = estimate of the variance of effect sizes; Chi^2 = test for heterogeneity; I^2 = percentage of heterogeneity across interventions; Z = effect size; df = overall indicator of the effect size on the meta-analysis results; *p* = statistically significant heterogeneity; green points = effect size of each study; black diamond for each variable = pooled effect size of the studies; final black diamond = subgroup effect size (Zhao et al., 2025; Rahimi et al., 2022; Hwang et al., 2012; Chu & Chang, 2014; Mohammed & Karpudewan, 2024; Khamparia et al., 2020; Lazarevic et al., 2024; Hwang et al., 2024; Wang & Zheng, 2021; García & Garzón, 2024; K. F. Chen et al., 2024; Hwang et al., 2013; Yang & Chang, 2013).

The self-efficacy subgroup consisted of n = 5 studies (Hwang et al., 2024; Khamparia et al., 2020; Lazarevic et al., 2024; Mohammed & Karpudewan, 2024; Wang & Zheng, 2021—digital games; Wang & Zheng, 2021—non-digital games), including 866 students in the EG and 855 in the CG. In this case, the intervention demonstrated a positive effect, with a combined MD of 0.72 (95% CI: 0.34–1.10; Z = 3.70; p = 0.0002), accompanied by high heterogeneity (I^2 = 93%). With respect to the academic performance subgroup, n = 9 studies were included (K. F. Chen et al., 2024; Chu & Chang, 2014; García & Garzón, 2024; Hwang et al., 2012, 2013, 2024; Khamparia et al., 2020; Yang & Chang, 2013; Zhao et al., 2025), comprising 261 participants in the EG and 259 in the CG. The combined effect was clearly

positive, with an MD of 4.46 (95% CI: 2.44–6.49; $Z = 4.32$; $p < 0.0001$), representing a large effect, together with moderate heterogeneity ($I^2 = 70\%$).

Finally, the results obtained across all variables showed an overall effect with a mean difference of 0.98 (95% CI: 0.68–1.29), indicating a global positive impact of GBL ($Z = 6.29$; $p < 0.0001$). Although motivation, self-efficacy, and academic performance are distinct constructs, a combined effect was reported as a global measure of GBL's impact on learning, considering that these results reflected interrelated aspects of performance and the educational experience. The dependency among outcomes was addressed using a random-effects model, which allowed for the incorporation of variability between studies and subgroups. The high heterogeneity values ($I^2 = 89\%$) showed that, while a general positive effect existed, the magnitude of the impact varied across different constructs and contexts, with the interventions' effectiveness remaining significant.

3.4. Risk of Bias and Quality Criteria of the Included Studies

Once the studies for the meta-analysis were selected, the reliability of the results was assessed using the *Cochrane Risk of Bias* tool, complemented with a Funnel Plot to detect the potential presence of publication bias. This combination allowed for an examination of the risk of bias in the psychological parameters studied—motivation, self-efficacy, and academic performance—within the context of Natural Sciences learning. The evaluation covered all three psychological domains, and as shown in Figure 3, the studies are distributed symmetrically around the vertical line corresponding to the null effect ($MD = 0$), indicating a low risk of bias across all assessed domains.

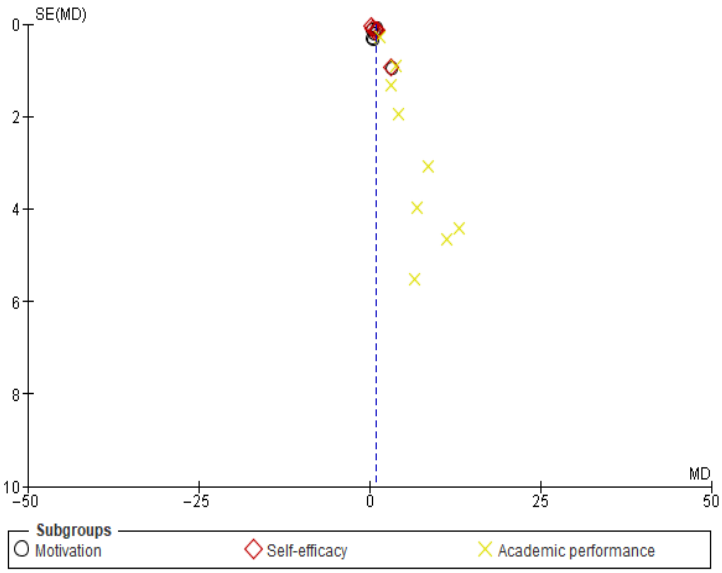

Figure 3. Overall risk of bias. Note: The citations related to each variable are presented as follows: motivation (Chu & Chang, 2014; Hwang et al., 2012; Khamparia et al., 2020; Mohammed & Karpudewan, 2024; Rahimi et al., 2022; Zhao et al., 2025), self-efficacy (Hwang et al., 2024; Khamparia et al., 2020; Lazarevic et al., 2024; Mohammed & Karpudewan, 2024; Wang & Zheng, 2021 [digital video games]; Wang & Zheng, 2021 [non-digital games]), and academic performance (K. F. Chen et al., 2024; Chu & Chang, 2014; García & Garzón, 2024; Hwang et al., 2012, 2013; Khamparia et al., 2020; Yang & Chang, 2013; Zhao et al., 2025).

Regarding motivation, the studies cluster closely and symmetrically, without significant deviations, suggesting a low risk of bias in this domain. Concerning self-efficacy, the studies are also positioned near the null effect line and show a symmetrical distribution, reinforcing the conclusion that the risk of bias in this domain is similarly low. Finally, for academic performance, although there is greater dispersion among the studies, no

pronounced asymmetry patterns are observed, indicating that, despite the variability, the risk of bias remains low in this domain.

3.5. Risk of Bias and Quality Criteria of the Included Studies

The findings of the meta-analysis were also evaluated using the GRADE system. The evidence for the motivation outcome was rated as low, mainly due to high inconsistency and moderate imprecision. Self-efficacy likewise showed low certainty, influenced by the very high heterogeneity across studies. In contrast, academic performance demonstrated moderate certainty of the evidence, supported by a consistent effect and lower imprecision. These results indicate that the evaluated interventions produced positive effects, although the strength of the evidence varied depending on the outcome analyzed (Table 4).

Table 4. Evidence summary of interventions on motivation, self-efficacy, and academic achievement.

Outcome	Studies	Effect (MD [95% CI])	Imprecision	Indirectness	Publication Bias	Certainty
Motivation	6	0.75 [0.33, 1.17]	Moderate	None	Low	Low
Self-efficacy	6	0.72 [0.34, 1.10]	Moderate	None	Low	Low
Academic achievement	9	4.46 [2.44, 6.49]	Low	None	Low	Moderate

Note. MD = Mean Difference; 95% CI = 95% Confidence Interval. Certainty reflects the overall confidence in the effect estimate according to GRADE criteria. Imprecision indicates the reliability of the effect estimate. Indirectness reflects how directly the evidence applies to the population, intervention, and outcomes studied. Publication Bias indicates the likelihood of selective publication.

3.6. Results of the Keyword Co-Occurrence Analysis

In the keyword co-occurrence analysis, three main thematic groups related to the use of games in educational contexts were identified (Figure 4). The analysis was conducted with a minimum keyword frequency of 10, which allowed for the selection of the most relevant terms. A total of 70 keywords met this criterion and were included in the analysis. Link strength thresholds were applied to filter out weaker connections, ensuring that the resulting clusters reflected only the most significant relationships between terms. These robust co-occurrence patterns facilitated the identification and interpretation of the three main thematic clusters, providing a comprehensive overview of the key associations in the literature.

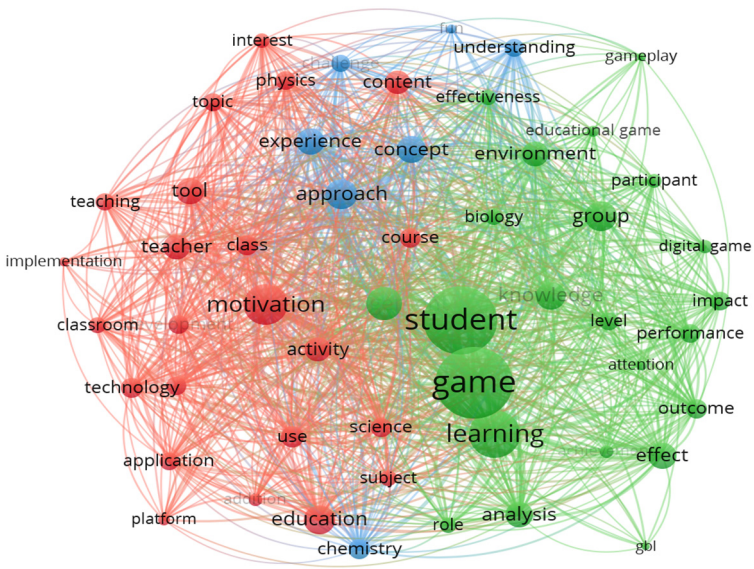

Figure 4. Thematic or cluster node map.

The visualization showed that these groups were strongly interconnected, indicating that the research did not address concepts in isolation but rather as interdependent elements within the teaching and learning process. In this regard, the identified groups were:

The first group (green) was directly related to student learning and outcomes. Prominent terms in this cluster included *student*, *learning*, *game*, *performance*, and *knowledge*. The presence of larger nodes in this group reflected that most studies focused on examining the impact of games on academic progress, knowledge acquisition, and participant performance.

The second group (red) was associated with the pedagogical context and conditions for implementing games in the educational environment. Terms such as *education*, *teacher*, *classroom*, *technology*, *motivation*, and *tool appeared* in this cluster. This suggested that a significant portion of the literature analyzed the role of teachers, the integration of technology, and motivation as key factors for incorporating games into educational practice.

The third group (blue) clustered concepts related to design and the pedagogical experience, with terms such as *concept*, *content*, *experience*, *approach*, and *effectiveness*. This group reflected studies aimed at evaluating the quality of the educational experience as well as the effectiveness of game-based approaches in enhancing content comprehension.

Overall, the results showed that previous research addressed the use of educational games from three complementary perspectives: their impact on student learning, the pedagogical conditions necessary for their implementation, and the design elements that influence the teaching experience.

4. Discussion

The results of the present study allow us to achieve the general objective of analyzing the effect of GBL on motivation, self-efficacy, and academic performance in the learning of Natural Sciences. Based on the qualitative analysis of the studies, GBL can be defined as the creation of learning environments that integrate digital or traditional games to enhance students' acquisition of knowledge and skills. Approaches based on video games or educational games have consistently demonstrated greater improvements in learning and knowledge retention compared to traditional methodologies, highlighting the integration of knowledge organization strategies, such as the concept map-embedded game approach (CM-DGBL), which significantly improves learning outcomes while reducing students' cognitive load (Hwang et al., 2013). From a theoretical perspective, these effects can be explained by cognitive load theory, which posits that instructional designs that reduce extraneous cognitive processing allow learners to use working memory efficiently for meaningful learning (Sweller, 1988). By scaffolding challenges and feedback within game environments, GBL reduces unnecessary cognitive burden and enhances germane cognitive processing, thereby improving retention and comprehension.

Furthermore, these systems promote deep learning behaviors, problem-solving awareness, and critical thinking tendencies, while digital game authoring (DGA) strengthens these skills and enhances academic performance (K. F. Chen et al., 2024; Hwang et al., 2024; Yang & Chang, 2013). At the motivational level, GBL consistently outperforms conventional methods, increasing overall motivation and improving key subcomponents such as confidence, satisfaction, self-efficacy, and the perception of intellectual accessibility of the content (García & Garzón, 2024; Cha et al., 2017; Hwang et al., 2013; Lazarevic et al., 2024; Sung & Hwang, 2013), improvements that appear to be directly related to long-term knowledge internalization. Therefore, the evidence suggests that these technological tools constitute relevant and complementary strategies to enhance academic performance, retention, and motivation in the disciplines of Natural Sciences: Biology, Geology, Physics, Chemistry, and Astronomy. This pattern aligns with *Self-efficacy Theory* (Bandura, 1994), which asserts that mastery experiences and supportive feedback enhance learners' beliefs in their capa-

bilities, leading to increased engagement and persistence. Furthermore, self-determination theory (Deci & Ryan, 1985) explains the motivational benefits of GBL by highlighting how autonomy, competence, and relatedness—core psychological needs—are satisfied through game mechanics, thereby fostering intrinsic motivation.

Science, technology, and GBL are closely interconnected in contemporary educational approaches, particularly in the Natural Sciences. Science is commonly described as the study of the natural world and the search for explanations for its phenomena and objects, while technology is linked to science through its focus on solving human-centered challenges. The application of GBL facilitates this integration by providing students with contextual cues that connect new learning with prior knowledge (Y. R. Lin, 2018). Similarly, analog and board games, such as *Solar System Traveler*, based on analogical concept-construction models, have been shown to improve learning outcomes, motivation, and knowledge acquisition in Astronomy education at the primary level (Cheng & Yeh, 2025). Escape Room-style activities also contribute to increased motivation and engagement in Natural Sciences learning (Naumoska et al., 2023). Providing domain-specific information before the gameplay phase, as suggested by Hu et al. (2024), can enhance both approach and avoidance motivation, aligning with Bandura's (1994) *Self-Concept and Self-Efficacy Theory* which posits that students who understand the learning context in advance feel more confident and motivated. Overall, GBL promotes motivation, self-efficacy, and academic performance among digital-native students by integrating interactive dynamics, immediate feedback, and progressively challenging tasks. This demonstrates that GBL engages students behaviorally, cognitively, and affectively by scaffolding learning experiences that promote self-regulated learning (Zimmerman, 2000), enhance mastery orientation, and support psychological needs identified by *Self-determination Theory*. By granting students autonomy and recognizing their achievements, GBL strengthens intrinsic motivation and self-efficacy, sustaining engagement and improving learning outcomes (Kuo et al., 2024). Additionally, GBL fosters the development of inquiry skills, making video games like *BioScientist* valuable tools for Biology education (Bónus et al., 2024; Srisawasdi & Panjaburee, 2019).

The findings of this review suggest that GBL acts as a catalyst for academic performance, motivation, and self-efficacy in the Natural Sciences across all educational levels. The results are consistent with active learning models, such as discovery, meaningful reception, and inquiry, as games appear to enable the exploration of concepts, integrate new information with prior knowledge, and solve problems interactively. Moreover, based on the reviewed literature, it can be concluded that beyond improving academic outcomes, GBL may also contribute to reducing cognitive load and strengthening affective and collaborative competencies, which are essential for long-term learning in Physics, Chemistry, Biology, Geology, and Astronomy. For example, in another line of research, in a meta-analysis of serious games, Wouters et al. (2013) found that educational games not only influence cognitive processes but also affect motivational and affective dimensions, and that learning gains were higher when learners worked in groups, highlighting the role of social interaction in enhancing learning outcomes.

In response to the first research question regarding the average effect size of game-based interventions on motivation, self-efficacy, and academic performance when comparing different EGs with CGs, our meta-analytic results highlight the significant value of incorporating GBL into the teaching and learning of Natural Sciences. The findings indicate that, regardless of the duration of the interventions, GBL consistently enhances students' motivation, self-efficacy, and academic performance. The EGs showed notable improvements across all analyzed variables, demonstrating that the use of games positively impacts learning outcomes. Furthermore, the risk of bias in the included studies was low, supporting the reliability and robustness of these results.

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Nonetheless, although the certainty of the evidence for motivation and self-efficacy was rated as low according to the GRADE approach, this primarily reflects the high heterogeneity among studies and the methodological limitations inherent to educational research, rather than the absence of a real effect. This heterogeneity is evident in the variability of effect sizes, differences in study designs, diverse sample characteristics, and the use of multiple measurement instruments, which can lead to variations in the observed impact of GBL interventions. Despite these differences, the effects on motivation and self-efficacy were statistically significant and of moderate magnitude, indicating genuine positive impacts. For practitioners, "low certainty" means that, although game-based interventions reliably improve motivation and self-efficacy, the specific magnitude of these effects may vary depending on contextual factors, such as age group, type of game, and instructional integration. In contrast, academic performance showed a large effect with moderate certainty, suggesting greater robustness. Overall, these findings support the value of GBL interventions.

On the other hand, in response to the second research question, the results obtained from the co-occurrence analysis suggest that the use of games in educational contexts is approached from a comprehensive perspective, where learning objectives, student motivation, and teachers' pedagogical decisions converge. The predominance of terms such as student, learning, and game indicates that research has primarily focused on the impact of games on learning and academic performance. In this regard, the identified thematic clusters help interpret the meta-analytic findings by linking the observed quantitative effects on motivation, self-efficacy, and academic performance with the dominant conceptual and methodological approaches in the literature.

However, the presence of concepts related to the teacher's role, technology, and classroom organization suggests that implementing these strategies requires specific structural and methodological conditions. Additionally, studies that explore the experience and effectiveness of the approach highlight that it is not only important to incorporate a game but also how it is designed and integrated with the content. Overall, these findings reflect that the incorporation of games in education is a complex process that demands a balance between technological innovation, instructional planning, and student support to achieve meaningful outcomes. The sensitivity analyses confirmed that the main conclusions remained consistent: the effect sizes and the direction of the results did not change significantly when considering possible variations in the included studies or in the analysis methods, supporting the robustness of our findings.

Within the framework of the four pillars of education proposed by the United Nations Educational, Scientific and Cultural Organization (UNESCO; Delors, 1996), the findings of this meta-analysis provide a solid basis for understanding how this methodology enhances learning in the natural sciences. First, GBL contributes to learning to know, as EG students show significant improvements in conceptual understanding and greater motivation to explore scientific phenomena, which are essential aspects of investigative thinking. Likewise, GBL promotes learning to do, as many games require applying the scientific method,

solving problems, and making informed decisions in simulated environments, strengthening practical and procedural skills. In relation to learning to live together, collaborative games promote teamwork, communication, and negotiation, which are fundamental social skills for collectively facing scientific and environmental challenges. Finally, GBL promotes learning to be, as it encourages autonomy, self-regulation, curiosity, and responsibility, which are key attributes for the comprehensive development of students. Overall, empirical evidence on GBL largely converges with UNESCO's educational vision, showing that playful environments not only enrich the teaching of natural sciences, but also contribute to a more comprehensive and meaningful education.

GBL can be conceptualized as a factor that enhances students' motivation and self-efficacy, in accordance with *Self-determination Theory* (Deci & Ryan, 1985) and *Self-Efficacy Theory* (Bandura, 1994). The increase in motivation and confidence, in turn, facilitates greater cognitive engagement and the application of higher-order thinking skills, consistent with the principles of active learning and inquiry-based learning. Moreover, this process aligns with UNESCO's four pillars of education, as GBL not only promotes "learning to know" and "learning to do" through problem-solving and conceptual understanding, but also supports "learning to live together" and "learning to be" by fostering collaboration, autonomy, and self-regulation. Therefore, GBL can act as a catalyst for both affective and cognitive development, ultimately leading to improved academic performance in the Natural Sciences.

GBL, through both physical and digital games, enhances meaningful learning and fosters active student engagement. The integration of information and communication technologies (ICT) supports the development of digital competencies and aligns with quality education, in accordance with the United Nations Sustainable Development Goals of the 2030 Agenda (Soriano-Sánchez et al., 2025). Furthermore, these tools ensure principles of universal accessibility, allowing for the accommodation of diverse learners and the removal of physical, cognitive, sensory, and emotional barriers, thereby contributing to an inclusive and equitable educational experience (Soriano-Sánchez, 2025). This approach uses a student-centered teaching model that addresses contemporary educational challenges by combining rigorous knowledge acquisition with playful and contextual methodologies, promoting deep understanding, practical application of content, and personalized learning.

4.1. Limitations of the Present Study and Future Research Directions

Despite the results obtained, this study presents certain limitations. First, the exclusive use of the WoS and Scopus databases may have led to the inadvertent omission of relevant studies published in other electronic sources or in the gray literature, which could affect the representativeness of the findings. Additionally, although rigorous selection and analysis criteria were applied, high heterogeneity persists across the set of interventions, possibly due to methodological differences among the included studies, such as variations in research designs and/or the instruments used to measure the variables. To address these limitations, future research should expand the search to other databases and repositories, allowing for the identification and inclusion of additional studies in subsequent meta-analyses. Furthermore, it would be advisable to prioritize studies that have measured the same variables using standardized instruments, enabling an assessment of whether these differences contribute to the observed heterogeneity.

As an additional future research direction, it would be valuable to explore the integration of emerging technologies within GBL environments. For example, incorporating AI-assisted help-seeking features, such as ChatGPT-style chatbots based on OpenAI's GPT-5.2 model, within digital GBL platforms—such as the video game Summon of Magicrystal—could support students in designing and navigating GBL environments, thereby enhancing science learning (C. H. Chen & Law, 2025). Investigating these innovative tools would

provide insights into their potential effects on motivation, self-efficacy, and academic performance, while avoiding overgeneralization beyond the empirical scope of current meta-analytic evidence. This approach would help determine the true effects of the interventions more accurately and improve the comparability and validity of the findings.

4.2. Practical Implications of the Results

The practical implications of this study highlight the importance of designing pedagogical methodologies that intentionally integrate GBL across different educational levels for the teaching and learning of Natural Sciences content, with the aim of enhancing students' motivation, self-efficacy, and academic performance. GBL emerges as an effective educational methodology, capable of combining technology and/or manipulative games, motivation, and meaningful learning, provided it is adapted to the context, student profiles, and their educational needs. These findings support the reflective and evidence-based incorporation of GBL practices, promoting more dynamic, inclusive learning environments oriented toward the holistic development of students, while also contributing to ensuring high-quality education aligned with the demands of contemporary society and the development of curricular competencies across all educational stages.

5. Conclusions

This study demonstrates that the integration of GBL methodologies significantly contributes to improving students' motivation, self-efficacy, and academic performance in the learning of Natural Sciences, provided it is implemented in a planned manner that simultaneously considers the teacher's role, student motivation, and the pedagogical design of the learning experience.

Meta-analytic findings confirm that GBL is an effective methodology for enhancing these variables across different educational levels. Its planned implementation represents an opportunity to create more dynamic and inclusive learning environments through a student-centered teaching model.

Keyword co-occurrence analysis shows that research on the use of games in educational contexts is organized around three interrelated axes: student learning outcomes, the pedagogical context of implementation, and the design of the learning experience. The strong connections between these groups indicate that GBL should not be understood as an isolated element but as an integrated strategy in which academic performance, teacher involvement, technology, motivation, and pedagogical design quality interact. These findings suggest that the effectiveness of games in education depends on the interaction of these components, reaffirming their potential as an innovative tool to enrich teaching and learning processes.

In sum, the implementation of GBL offers an opportunity to transform the learning experiences of digital-native students, aligning with their natural ways of interacting with information. By promoting active participation, a sense of achievement, and autonomy, this methodology strengthens both motivation and self-efficacy, translating into improved academic performance. In this way, GBL establishes itself as an effective pedagogical approach for addressing the educational challenges of the 21st century in the teaching of Natural Sciences, contributing to improved educational quality and supporting the achievement of the Sustainable Development Goals outlined in the 2030 Agenda.

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